

KENDAL CARMONA HERRERA

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Professional Summary

Full Stack Engineer with 5+ years of experience building enterprise SaaS platforms, AI-powered automation systems, and conversational commerce solutions using Angular, NestJS, TypeScript, MongoDB, PostgreSQL, and cloud-native technologies. Experienced in performance optimization, clean and hexagonal architecture, multi-tenant systems, SAP Business One integrations, and production-grade AI workflows. Proven track record of reducing application load times from ~50 s to under 5 s, improving application performance by 70%+, reducing codebase size by 82%, and delivering 700+ automated tests with over 90% coverage on critical business flows. Skilled in Agile environments, technical decision-making, engineering standards, and scalable software delivery.

Technical Skills

- **Frontend:** Angular 14–21, TypeScript, JavaScript, RxJS, Signals, Standalone Components, OnPush, HTML5, CSS3, SCSS, PrimeNG, PrimeReact, React, Next.js, Tailwind CSS, PWA, IndexedDB, Service Workers.
- **Backend:** NestJS, Node.js, Express, REST APIs, Webhooks, WebSocket, Socket.IO, Swagger/OpenAPI, DTOs, Guards, Interceptors, Pipes.
- **Databases:** PostgreSQL (GIN indexing, RLS, CDC, RPCs, query optimization), Supabase, MongoDB, Mongoose, SQL Server, SAP HANA, migrations.
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Nginx, Bitbucket Pipelines, Git, Jira
- **AI & Automation:** n8n, OpenAI API, Anthropic, OpenRouter, Claude Code, Codex, MCP (Model Context Protocol), prompt engineering.
- **Testing & QA:** Playwright, Jest, TDD, E2E testing, unit testing, test automation (700+ tests shipped).
- **Architecture:** Clean Architecture, Hexagonal Architecture, SOLID, Repository Pattern, Facade & Strategy Patterns, multi-tenancy, event-driven design.
- **Methodologies:** Agile (Scrum/Kanban), code review, pair programming, technical documentation, debugging.
- **Integrations:** SAP Business One (sales orders, quotes, manufacturing orders, goods receipts/deliveries), Service Layer, Evolution API, WhatsApp Business API, PDFmake, ExcelJS.
- **Languages:** Spanish (Native), English (B1+ — Intermediate, actively improving)
- **Continuous Learning:** Actively expanding knowledge in Python and .NET fundamentals while reinforcing cloud and infrastructure skills with AWS, Kubernetes, and Docker in modern application environments.
- **Collaboration & Leadership:** Technical decision-making, cross-functional collaboration, code review, pair programming, problem-solving, technical documentation, Agile teamwork.

Professional Experience

Full Stack Engineer

RDS – Costa Rica | 2021 – Present

Enterprise SaaS Platform Engineering & Performance Optimization

- Architected and led the redesign of a mission-critical multi-tenant enterprise platform module, reducing codebase size by 82% through a modular Angular and NestJS architecture using standalone components, lazy loading, and focused services.
- Engineered frontend and database performance optimizations using pg_trgm GIN indexing, query tuning, cache-first strategies, and debounce optimization, reducing search and interaction latency from 3–5 s to under 1 s.
- Reduced critical application page load time from ~50 s to under 5 s through lazy loading, OnPush change detection, and backend query optimization.

- Designed and implemented a scalable RBAC system with tenant isolation, permission-based routing, and backend authorization guards for secure multi-tenant SaaS environments.
- Reworked authentication flows and data models to support identity migration and guard standardization aligned with enterprise security requirements.
- Delivered audit logging and database-level business rule enforcement to improve data integrity, traceability, and operational reliability.
- Developed offline-first workflows using Service Workers and IndexedDB to support low-connectivity enterprise environments.
- Built and maintained 700+ automated tests using Playwright and Jest across unit and E2E layers, increasing coverage above 90% on critical business flows.
- Diagnosed and resolved production incidents involving authentication, synchronization, messaging delivery, and distributed state consistency.
- Refactored a large-scale module from monolithic components into a modular and testable architecture with 150+ automated tests and clear separation of concerns.

SAP Business One Integration

- Designed and integrated SAP Business One workflows covering sales orders, quotations, manufacturing orders, goods receipts, and delivery processes.
- Built secure API layers that replaced direct ERP database access from automation services with validated and authenticated endpoints.

AI Automation Engineer & Conversational Systems

- Designed and deployed an AI-powered sales orchestration platform for WhatsApp commerce using n8n, OpenAI, Anthropic, PostgreSQL memory persistence, and state-machine-driven workflows.
- Automated inventory queries, payment processing flows, shipping coordination, and human escalation pipelines for conversational retail operations.
- Built an orchestrator-based architecture that centralized decision-making into deterministic workflows, improving reliability and reducing hallucination risk in production LLM pipelines.
- Implemented anti-hallucination safeguards, deterministic validation strategies, and defensive JSON parsing patterns for production AI workflows.
- Diagnosed and resolved issues related to conversational state management, payment proof validation, message routing, and workflow consistency in AI-driven systems.

Key Engineering Achievements

- Reduced enterprise platform codebase size by 82% through Angular and NestJS architecture modernization.
- Achieved 70%+ improvement in search and interaction performance through indexing strategies, query optimization, and frontend caching techniques.
- Reduced critical application load times from ~50 s to under 5 s using lazy loading, OnPush rendering, and backend performance tuning.
- Delivered 700+ automated tests across unit and E2E layers with over 90% coverage on critical workflows.
- Added 150+ automated tests during a large-scale WhatsApp module refactor.
- Built 20+ enterprise reporting templates using PDFmake and ExcelJS.
- Integrated SAP Business One across 4+ ERP operational flows including sales, manufacturing, and logistics processes.

Education

Bachelor's Degree in Information Technology Engineering
 Universidad Técnica Nacional, Costa Rica